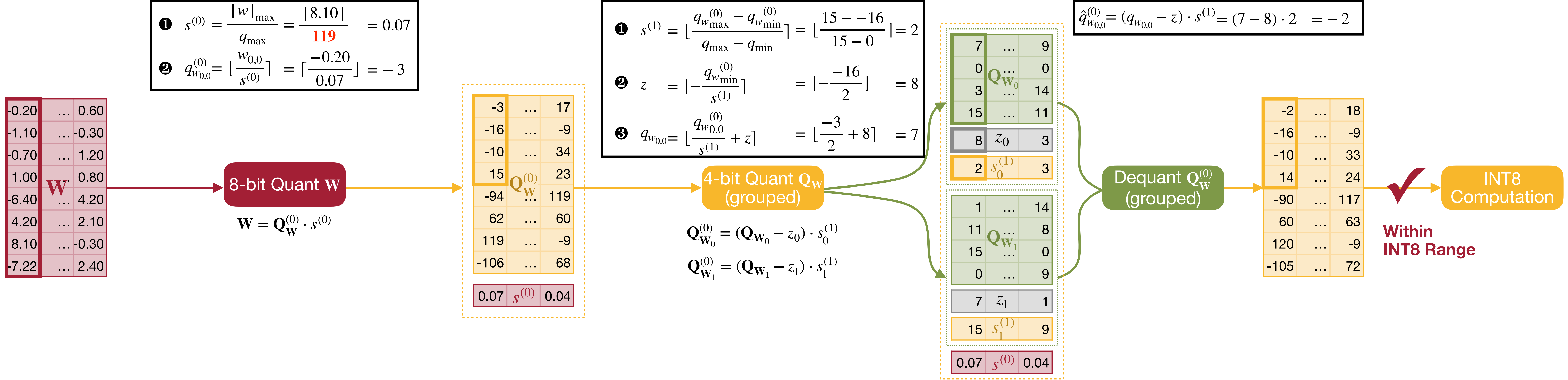

$s^{(0)} = \frac{|w|_{\max}}{q_{\max}} = \frac{|8.10|}{119} = 0.07$
$q^{(0)}_{w_{0,0}} = \lfloor \frac{w_{0,0}}{s^{(0)}} \rceil = \lceil \frac{-0.20}{0.07} \rfloor = -3$
8-bit Quant W
$\mathbf{W} = \mathbf{Q}^{(0)}_{\mathbf{W}} \cdot s^{(0)}$
$s^{(1)} = \lfloor \frac{q^{(0)}_{w\max} - q^{(0)}_{w\min}}{q_{\max} - q_{\min}} \rceil = \lfloor \frac{15 - -16}{15 - 0} \rceil = 2$
$z = \lfloor - \frac{q^{(0)}_{w\min}}{s^{(1)}} \rceil = \lfloor - \frac{-16}{2} \rceil = 8$
$q_{w_{0,0}} = \lfloor \frac{q^{(0)}_{w0,0}}{s^{(1)}} + z \rceil = \lfloor \frac{-3}{2} + 8 \rceil = 7$
4-bit Quant $\mathbf{Q}_\mathbf{W}$ (grouped)
$\mathbf{Q}^{(0)}_{\mathbf{W}_0} = (\mathbf{Q}_{\mathbf{W}_0} - z_0) \cdot s^{(1)}_0$
$\mathbf{Q}^{(0)}_{\mathbf{W}_1} = (\mathbf{Q}_{\mathbf{W}_1} - z_1) \cdot s^{(1)}_1$
$\hat{q}^{(0)}_{w_{0,0}} = (q_{w_{0,0}} - z) \cdot s^{(1)} = (7 - 8) \cdot 2 = -2$
Dequant $\mathbf{Q}^{(0)}_\mathbf{W}$ (grouped)
Within INT8 Range
INT8 Computation